

Representation transfer for GFP fluorescence prediction

AI Research Intern assignment

1. Task

The goal is to predict GFP fluorescence (DMS_score) for single amino-acid variants of avGFP, and check whether a protein model from a different family adds anything on top of an ESM base model. The split is grouped by mutated position, so no position in training appears in test. This tests generalization to sites not seen in training, not just fitting the scan. Grading is Spearman rank correlation on the 213 test variants, with MSE reported as well.

2. Models and features

Base model: ESM-2 35M (facebook/esm2_t12_35M_UR50D), a masked model. Second model: ProtGPT2 (nferruz/ProtGPT2), an autoregressive GPT-2 style model, so a different family from ESM. I originally planned to use ProGen2-small, but its custom model code did not load correctly with the transformers version available on Colab (a get_head_mask error, then tensors stuck on a meta device that would not move to the GPU). ProtGPT2 uses the standard GPT-2 architecture with no custom code, so it loads cleanly and still gives an autoregressive model to compare against ESM. Both encoders are kept frozen, since finetuning is out of reach on the available compute. For each sequence I take the last hidden layer and mean pool over the tokens to get one vector per model (480-dim for ESM, 1280-dim for ProtGPT2). Features are cached so the encoders only run once.

3. Method

The prediction head is ridge regression in every case, so the comparison is about the features, not the head. Features are standardized using training statistics only. Four setups are compared: ESM only (the main baseline), ProtGPT2 only, the two vectors concatenated, and a transfer method.

The transfer method follows the idea in the SoupFold paper of learning a mapping between two models' representations. I train a small MLP (two hidden layers of 512, ReLU) to predict the ESM vector from the ProtGPT2 vector, using MSE on the training set. I then average the real ESM vector with the mapped ProtGPT2 vector and fit ridge on that average. Averaging in ESM space is the soup step: the mapped ProtGPT2 vector acts like a second view of the same sequence, expressed in the base model's space.

4. Fitting rules

All learned parts (scaler, mapper, ridge) are fit on the 657 training examples during development. The 214 validation examples are used only to pick the ridge alpha for each method. After the settings are fixed, every learned part is reinitialized and refit on the combined 871 train+val examples, and the 213 test variants are scored once. Test labels are used for the final metric only, never for fitting or selection.

5. Validation results

Selected on the 214 validation variants.

Method	Alpha	Val Spearman	Val MSE
ESM only	10	0.137	1.798

ProtGPT2 only	10	0.136	0.647
Concat (ESM+ProtGPT2)	1	0.179	1.353
Transfer (mapped + soup)	0.1	0.243	1.404

Mapper quality: the MLP reached a validation R2 of -0.012 at reproducing the ESM vector from the ProtGPT2 vector, meaning it did not explain ESM's variation better than just predicting the mean. Even so, averaging its output with the real ESM vector gave the best validation Spearman of the four methods, so the mapped vector still contributed some useful signal on top of the raw ESM features, even though it is a poor reconstruction of them.

6. Final test results

Refit on 871 train+val, scored once on the 213 test variants. MSE is in original DMS_score units.

Method	Test Spearman	Test MSE
ESM only (baseline)	0.119	0.455
Concat (ESM+ProtGPT2)	0.300	4.054
Transfer (mapped + soup)	0.235	0.438

7. What the results show

Both concat and transfer beat the ESM-only baseline on test Spearman (0.300 and 0.235 versus 0.119), so adding ProtGPT2 information did help ranking on held-out positions in this run. The two methods disagree on MSE though. Concat has the best Spearman but its MSE (4.054) is almost ten times higher than the ESM baseline, which means its actual predicted values are far off even though the ranking improved, most likely because concatenating the full ESM and ProtGPT2 vectors (1760 dimensions total) with a fairly small alpha of 1 overfits the 657 training points and produces predictions with the wrong scale. Transfer keeps MSE close to the ESM baseline (0.438 versus 0.455) while still improving Spearman by about double, which is a more balanced result and looks like the more trustworthy method between the two. The pattern is consistent with the point in the assignment brief that improvement is a research question and not guaranteed: adding the second model changed the ranking, but concat's exploding MSE shows that not every way of combining the two representations is safe to use as is.

8. Uncertainties

Every variant here differs from the same reference by a single residue, so ESM and ProtGPT2 see almost identical inputs (splits are about 99% identical to each other). This makes it somewhat surprising that ProtGPT2 features added a usable signal at all, and it is worth checking with a different random seed or a different alpha grid whether the improvement holds up, since the training set is only 657 to 871 examples. The mapper's negative R2 suggests the ESM-to-ProtGPT2 mapping is not learning a strong shared structure, so the improvement from transfer may partly come from the averaging acting as a mild regularizer rather than from a genuinely useful cross-model mapping. Concat's large MSE is a caution that Spearman alone is not a full picture of the model's quality. With one protein and one assay, these numbers should not be read as a general claim about masked versus autoregressive protein models.

Reproducing: the notebook `gfp_transfer.ipynb` in the repo runs end to end on Colab with a GPU. Checkpoints are `facebook/esm2_t12_35M_UR50D` and `nferruz/ProtGPT2` (standard transformers classes, no custom code needed). It prints every number in the tables above and saves a prediction CSV for each final method.